Lecture 14

The Lorentz group and its representations

Objectives. *Spacetime now, with $\hbar = c = 1$. From the constancy of the speed of light we extract the Minkowski interval and its symmetry group $\mathrm{O}(1,3)$, then dissect it: four components, boosts and rapidity, and the polar anatomy $\Lambda = (\text{rotation})(\text{boost})$, which makes the restricted group connected but — Lecture 11's loop travelling along — not simply connected. The Lie algebra's surprise is a sign, $[K,K] = -L$: boosts do not close, and the Thomas–Wigner rotation is the debris. Complexifying, the algebra falls apart into* two *commuting copies of $\mathfrak{sl}(2,\mathbb{C})$, and Lecture 10 classifies everything: irreducibles (j_1, j_2). The double cover $\mathrm{SL}(2,\mathbb{C}) \to \mathrm{SO}^+(1,3)$ — Lecture 11 one signature up — is proved simply connected by polar decomposition, and Lecture 9's lifting theorem is finally spent in earnest: every (j_1, j_2) integrates. Weyl spinors, Dirac spinors, and four-vectors take their seats. The sharpest theorem is negative — no nontrivial finite-dimensional* unitary *representations — and its moral organises the finale: fields transform finitely and non-unitarily, states unitarily and infinitely. A starred section lets the Dirac equation assemble itself from intertwiners.*

14.1 Minkowski space from physics

Two postulates. First, *relativity*: inertial observers exist, and the coordinate transformations between them carry uniform straight-line motion to uniform straight-line motion — which forces them to be affine (a classical theorem of affine geometry that we state and use), and, after matching origins, linear. Second, *light*: one and the same speed $c = 1$ in every inertial frame. On $\mathbb{R}^4$ with coordinates $x = (x^0, \vec{x}\,)$, the second postulate says exactly that every transformation Λ preserves the *light cone* $\{x : \eta(x,x) = 0\}$ of the quadratic form

$$\eta(x,y) = x^0 y^0 - \vec{x}\cdot\vec{y} = x^{\mathsf{T}} \eta\, y, \qquad \eta = \mathrm{diag}(1,-1,-1,-1).$$

This derivation uses two stated inputs. The first is the affine-geometry theorem just named. The second takes us from the cone to the full form; it is elementary but fiddly.

Lemma 14.1 (From the cone to the interval; stated)**.** *A linear bijection of $\mathbb{R}^4$ mapping the light cone onto itself satisfies $\Lambda^{\mathsf{T}}\eta\Lambda = \lambda\,\eta$ for some $\lambda > 0$.*

The leftover scale dies by reciprocity: λ can depend only on the relative speed of the frames (isotropy), the inverse transformation has the same relative speed, so $\lambda(\Lambda) = \lambda(\Lambda^{-1})$, while $\lambda(\Lambda)\lambda(\Lambda^{-1}) = \lambda(\mathbb{1}) = 1$; hence $\lambda = 1$. The symmetries of special relativity are therefore the linear maps preserving η on the nose,

$$\mathrm{O}(1,3) = \{\Lambda \in \mathrm{GL}(4,\mathbb{R}) : \Lambda^{\mathsf{T}}\eta\Lambda = \eta\},$$

a closed subgroup of $\mathrm{GL}(4,\mathbb{R})$ and so a matrix Lie group — the last cage of Lecture 8's zoo to be opened. (Translations preserve intervals too; the full affine symmetry group, the Poincaré group, is Lecture 15's. Today the homogeneous part has work enough.)

14.2 The structure of the Lorentz group

Two scalars split the group in four. Taking determinants of $\Lambda^{\mathsf{T}}\eta\Lambda = \eta$ gives $\det\Lambda = \pm 1$. The $(0,0)$ entry of the same identity reads

$$(\Lambda^0{}_0)^2 = 1 + \sum_i (\Lambda^i{}_0)^2 \;\geq\; 1,$$

so $\Lambda^0{}_0 \geq 1$ or $\Lambda^0{}_0 \leq -1$: a Lorentz transformation either preserves or reverses the direction of time, never hesitates.

Proposition 14.2 (Four pieces)**.** *The map* $\Lambda \mapsto (\det\Lambda, \operatorname{sgn}\Lambda^0{}_0)$ *is a homomorphism* $\mathrm{O}(1,3) \to \{\pm1\}\times\{\pm1\}$*, surjective, and its four fibres are the connected components of* $\mathrm{O}(1,3)$*. The kernel*

$$\mathrm{SO}^{+}(1,3) = \{\Lambda : \det\Lambda = 1,\ \Lambda^0{}_0 \geq 1\}$$

— the restricted Lorentz group *— is the identity component; coset representatives for the other three are the parity* $P = \mathrm{diag}(1,-1,-1,-1)$ *(the metric itself, by a small joke of the formalism), time reversal* $T = \mathrm{diag}(-1,1,1,1)$*, and* $PT = -\mathbb{1}$*.*

Figure 14.1 is the proposition at a glance: a 2×2 table of components, the restricted group $\mathrm{SO}^{+}(1,3)$ holding the identity in one corner, the discrete operations P, T, PT carrying it to the other three.

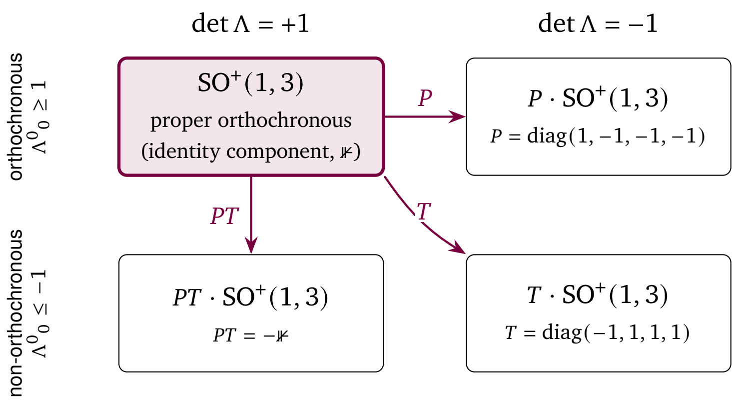

Figure 14.1: The four connected components of the Lorentz group $\mathrm{O}(1,3)$, separated by the two signs that no continuous path can change: the determinant $\det\Lambda = \pm1$ and the orientation of time $\Lambda^0{}_0 \geq 1$ versus ≤ -1. Only the proper orthochronous component $\mathrm{SO}^{+}(1,3)$ contains the identity and is reached by exponentiating $\mathfrak{so}(1,3)$; it is the group whose representation theory Lecture 14 develops. The other three are its cosets, reached from it by the discrete operations parity P, time reversal T, and their product $PT = -\mathbb{1}$. The Lie algebra sees the identity component only — the discrete symmetries must be supplied by hand.

Proof. Determinants multiply; the content is that $\operatorname{sgn}\Lambda^0{}_0$ does. From $\Lambda^{-1} = \eta\Lambda^{\mathsf{T}}\eta \in \mathrm{O}(1,3)$ the *rows* obey the same identity as the columns, so $\sum_i (\Lambda^0{}_i)^2 = (\Lambda^0{}_0)^2 - 1$. Cauchy–Schwarz then bounds the cross term in $(\Lambda\Lambda')^0{}_0 = \Lambda^0{}_0\Lambda'^0{}_0 + \sum_i \Lambda^0{}_i\Lambda'^i{}_0$:

$$\Big|\sum_i \Lambda^0{}_i\Lambda'^i{}_0\Big| \le \sqrt{(\Lambda^0{}_0)^2-1}\,\sqrt{(\Lambda'^0{}_0)^2-1} < |\Lambda^0{}_0|\,|\Lambda'^0{}_0|,$$

so the sign of $(\Lambda\Lambda')^0{}_0$ is the product of the signs: a homomorphism. Surjectivity: evaluate on $\not{1}, P, T, PT$. The four fibres are separated — both invariants are continuous with values in $\{\pm1\}$ — and each is connected because each is a translate of $\mathrm{SO}^+(1,3)$, whose connectedness is Corollary 14.4 below. ■

What lives inside $\mathrm{SO}^+(1,3)$? Rotations, embedded as $1 \oplus R$ with $R \in \mathrm{SO}(3)$ — they preserve x^0 and the Euclidean length, hence η. And *boosts*: the transformations mixing time with one spatial direction. Preserving $(x^0)^2 - (x^3)^2$ with determinant one and $\Lambda^0{}_0 \ge 1$ forces the hyperbolic rotation

$$B_3(\chi) = \begin{pmatrix} \cosh\chi & 0 & 0 & \sinh\chi \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \sinh\chi & 0 & 0 & \cosh\chi \end{pmatrix}, \qquad B_3(\chi)\,B_3(\chi') = B_3(\chi+\chi'), \tag{14.1}$$

the composition law from the hyperbolic addition formulas: a one-parameter group, with the *rapidity* χ as its honest additive parameter. The static worldline $\vec{x} = 0$ is carried to one of velocity $v = \tanh\chi$, so composing collinear boosts adds rapidities while velocities compose as

$$v'' = \tanh(\chi+\chi') = \frac{v+v'}{1+vv'}\ : \tag{14.2}$$

Einstein's addition law is the tanh addition formula, and no amount of composing escapes $(-1,1)$ — the speed of light is the boundary of a group orbit.

Rotations and boosts are not merely examples; they are the whole anatomy.

Theorem 14.3 (Polar anatomy of the Lorentz group)**.** *Every* $\Lambda \in \mathrm{SO}^+(1,3)$ *factors* uniquely *as*

$$\Lambda = R\,B, \qquad R = 1 \oplus \widetilde{R},\ \widetilde{R} \in \mathrm{SO}(3), \qquad B \text{ symmetric positive-definite,}$$

and the symmetric positive-definite elements of $\mathrm{SO}^+(1,3)$ *are exactly the* pure boosts e^W, $W = \left(\begin{smallmatrix} 0 & \vec{w}^{\mathsf T} \\ \vec{w} & 0_3 \end{smallmatrix}\right)$ *— the boost B_3 of (14.1) along e_3 when $\vec{w} = \chi e_3$, and its rotated copies in general.*

Proof. Sheet 8's polar decomposition gives $\Lambda = OS$ with O orthogonal and $S = \sqrt{\Lambda^{\mathsf T}\Lambda}$ symmetric positive-definite, uniquely. Everything to prove is that both factors are Lorentz. Set $M = \Lambda^{\mathsf T}\Lambda$; since $\Lambda^{\mathsf T} = \eta\Lambda^{-1}\eta$,

$$M = \eta\Lambda^{-1}\eta\Lambda, \qquad M\eta M = \eta\Lambda^{-1}\eta(\Lambda\Lambda^{-1})\eta\Lambda = \eta\ :$$

$M \in \mathrm{O}(1,3)$, and with $M^{\mathsf T} = M$ this reads $\eta M\eta = M^{-1}$. Now pass to the square root: $T = \eta S\eta$ is symmetric positive-definite with $T^2 = \eta M\eta = M^{-1} = (S^{-1})^2$, and positive square roots are unique (Sheet 8), so $\eta S\eta = S^{-1}$, i.e. $S \in \mathrm{O}(1,3)$. Positivity gives $S^0{}_0 = \langle e_0, Se_0\rangle > 0$ and $\det S > 0$, so $S \in \mathrm{SO}^+(1,3)$. Then $O = \Lambda S^{-1}$ lies in $\mathrm{SO}^+(1,3) \cap \mathrm{O}(4)$: from $O^{-1} = O^{\mathsf T}$ and $O^{\mathsf T} = \eta O^{-1}\eta$ we get $O\eta = \eta O$, and a matrix commuting with η is block diagonal, $(\pm1)\oplus\widetilde{R}$; orthochronicity selects $+1$ and the determinant puts $\widetilde{R}$ in $\mathrm{SO}(3)$. Finally the shape of S: write $S = e^W$ with $W = \log S$ symmetric (spectral calculus, Sheet 8); applying log to $\eta S\eta = S^{-1}$ gives $\eta W\eta = -W$, and a symmetric matrix anticommuting with η has vanishing diagonal blocks: $W = \left(\begin{smallmatrix} 0 & \vec{w}^{\mathsf T} \\ \vec{w} & 0 \end{smallmatrix}\right)$. For $\vec{w} = \chi e_3$ the exponential series collapses onto cosh and sinh in the $(0,3)$ block, reproducing (14.1); a rotation conjugates the axis anywhere. ■

Corollary 14.4 (Topology of the restricted group)**.** *The map* $(\widetilde{R}, \vec{w}) \mapsto (1 \oplus \widetilde{R})\, e^{W(\vec{w})}$ *is a homeomorphism* $\mathrm{SO}(3) \times \mathbb{R}^3 \to \mathrm{SO}^+(1,3)$ *(the polar factors depend continuously on* Λ*). Hence* $\mathrm{SO}^+(1,3)$ *is connected — justifying Proposition 14.2 — but* not *simply connected: a loop in a product contracts precisely when both factor loops do, and the loop* $t \mapsto R_z(2\pi t)$*, proved non-contractible in Lecture 11, travels along into* $\mathrm{SO}^+(1,3)$*.*

The full rotation that embarrassed SO(3) embarrasses spacetime too; the cure, as before, will be a double cover — but first, the algebra.

14.3 The Lie algebra $\mathfrak{so}(1,3)$

Differentiating curves $e^{tW} \in \mathrm{O}(1,3)$ at $t = 0$ — Lecture 8's calculation for $\mathrm{O}(n)$, with η inserted — gives

$$\mathfrak{so}(1,3) = \{W \in \mathfrak{gl}(4,\mathbb{R}) : W^{\mathsf{T}}\eta + \eta W = 0\}.$$

Writing W in 1+3 blocks, the condition says: vanishing diagonal corners, equal off-diagonal blocks ($\vec{w}$ and $\vec{w}^{\mathsf{T}}$), and an antisymmetric spatial block. So $\dim \mathfrak{so}(1,3) = 6$, with basis the three *rotation generators* $L_a = 0 \oplus L_a$ (Lecture 8's $\mathfrak{so}(3)$ matrices, parked in the spatial block — we recycle the name) and the three *boost generators* K_a with entries $(K_a)_{0i} = (K_a)_{i0} = \delta_{ai}$, the matrices found in Theorem 14.3. Their brackets are the lecture's turning point:

$$[L_a, L_b] = \epsilon_{abc} L_c, \qquad [L_a, K_b] = \epsilon_{abc} K_c, \qquad [K_a, K_b] = -\epsilon_{abc} L_c. \tag{14.3}$$

The first is Lecture 8; the second says $\vec{K}$ is a vector under rotations; the third is special relativity. Two entrywise checks settle the new ones — $[L_3, K_1]$: the only surviving products are $(L_3K_1)_{20} = (L_3)_{21} = 1$ and $(K_1L_3)_{02} = (L_3)_{12} = -1$, so $[L_3, K_1] = E_{20} + E_{02} = K_2$; and $[K_1, K_2] = E_{12} - E_{21} = -L_3$ — the rest follow by relabelling axes.

The minus sign in $[K, K] = -\epsilon L$ means *boosts do not close*: composing two non-collinear boosts is never a boost. By the polar anatomy it is a boost times a rotation, and that rotation by-product is the *Thomas–Wigner rotation* — physical, measurable, and responsible for the famous factor in the spin–orbit coupling of atomic electrons (Thomas precession). Sheet 14 computes the exact angle for perpendicular boosts, and Notebook 14 plots it. (With $+\epsilon L$ instead, (14.3) would be $\mathfrak{so}(4)$, the compact sibling; one sign separates Euclid from Einstein.) Note also, for later, an asymmetry: in the defining representation $\mathrm{i}L_a$ is Hermitian but $\mathrm{i}K_a$ is *anti*-Hermitian — remember it; in Section 14.5 it grows into a theorem.

Real-linear combinations cannot disentangle (14.3); complex ones can, and this is the lecture's central trick.

Theorem 14.5 (The great decoupling)**.** *In the complexification* $\mathfrak{so}(1,3)_{\mathbb{C}}$*, set*

$$A_a = \tfrac{1}{2}\big(L_a - \mathrm{i}K_a\big), \qquad B_a = \tfrac{1}{2}\big(L_a + \mathrm{i}K_a\big).$$

Then

$$[A_a, A_b] = \epsilon_{abc} A_c, \qquad [B_a, B_b] = \epsilon_{abc} B_c, \qquad [A_a, B_b] = 0,$$

and $L_a = A_a + B_a$*,* $K_a = \mathrm{i}(A_a - B_a)$*, so the* A*'s and* B*'s together span. Hence*

$$\mathfrak{so}(1,3)_{\mathbb{C}} \;\cong\; \mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C}) : \tag{14.4}$$

two commuting copies of the angular momentum algebra.

Proof. Expand, using (14.3) and $[K_a, L_b] = \epsilon_{abc}K_c$:

$$[A_a, A_b] = \tfrac{1}{4}\big(\epsilon_{abc}L_c - 2\mathrm{i}\,\epsilon_{abc}K_c + \epsilon_{abc}L_c\big) = \epsilon_{abc}\,\tfrac{1}{2}(L_c - \mathrm{i}K_c) = \epsilon_{abc}A_c,$$

the $[K,K]$ term contributing $+\epsilon L$ through $(-\mathrm{i})^2(-\epsilon L)$; likewise for B. In the cross bracket the K-terms cancel and the $[L,L]$, $[K,K]$ contributions kill each other: $[A_a, B_b] = \frac{1}{4}(\epsilon L + \mathrm{i}\epsilon K - \mathrm{i}\epsilon K - \epsilon L)_c = 0$. The displayed inversions show the six elements A_a, B_a span the six-dimensional $\mathfrak{so}(1,3)_{\mathbb{C}}$, so each triple spans a three-dimensional subalgebra and the sum is direct by dimension count. A complex algebra with basis obeying $\mathfrak{su}(2)$'s structure constants is a copy of $\mathfrak{su}(2)_{\mathbb{C}} = \mathfrak{sl}(2,\mathbb{C})$ (Lecture 10's complexification lemma, read backwards). ∎

Everything Lecture 10 built now applies twice over. The bookkeeping for a direct sum is a theorem worth proving once, properly.

Scope signal. The classification below is algebraic, finite-dimensional, and complex-linear. After integration it gives continuous finite-dimensional representations of $\mathrm{SL}(2,\mathbb{C})$, generally nonunitary. These are field transformation types. It is not a classification of unitary one-particle Hilbert-space representations; Lecture 15 changes to that category explicitly.

Theorem 14.6 (Representations of a direct sum)**.** *The finite-dimensional irreducible complex-linear representations of $\mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$ are exactly the outer tensor products*

$$\pi_{(j_1,j_2)}(X,Y) = \pi_{j_1}(X) \otimes \mathbb{1} + \mathbb{1} \otimes \pi_{j_2}(Y) \quad \text{on} \quad V_{j_1} \otimes V_{j_2},$$

labelled by pairs $(j_1, j_2) \in \big(\tfrac{1}{2}\mathbb{Z}_{\geq 0}\big)^2$, of dimension $(2j_1+1)(2j_2+1)$, pairwise inequivalent.

Proof. Write $\mathfrak{g}_1, \mathfrak{g}_2$ for the two summands. Let (π, V) be irreducible. Restricted to $\mathfrak{g}_1$, V is completely reducible into spin multiplets (complex-linear representations of $\mathfrak{sl}(2,\mathbb{C})$ are representations of $\mathfrak{su}(2)$, integrated along the simply connected $\mathrm{SU}(2)$ and decomposed by Lectures 10 and 12). Any operator T commuting with $\pi(\mathfrak{g}_1)$ maps a spin-j_1 submodule W to a quotient of itself — $T|_W$ intertwines — hence into the spin-j_1 isotypic component; since $\pi(\mathfrak{g}_2)$ commutes with $\pi(\mathfrak{g}_1)$, every isotypic component is invariant under the whole algebra, and irreducibility leaves exactly one: V is isotypic of some type j_1. Now let $M = \mathrm{Hom}_{\mathfrak{g}_1}(V_{j_1}, V)$ and consider the evaluation map $V_{j_1} \otimes M \to V$, $v \otimes T \mapsto Tv$: it intertwines the $\mathfrak{g}_1$-actions, is surjective (each irreducible summand is the image of its own inclusion), and matches dimensions by Schur, hence is an isomorphism. For $Y \in \mathfrak{g}_2$, composition $T \mapsto \pi(Y) \circ T$ defines $\sigma(Y)$ on M, and under evaluation $\pi(Y) = \mathbb{1} \otimes \sigma(Y)$; σ is a representation, and it is irreducible because a σ-invariant subspace N would hand V the invariant subspace $V_{j_1} \otimes N$. By Lecture 10, $\sigma \cong \pi_{j_2}$ for some j_2, and $\pi \cong \pi_{(j_1,j_2)}$. Conversely each $\pi_{(j_1,j_2)}$ is irreducible: an operator commuting with the $\mathfrak{g}_1$-half is $\mathbb{1} \otimes S$ by the same commutant computation, and commuting also with $\mathbb{1} \otimes \pi_{j_2}(\mathfrak{g}_2)$ makes S scalar (Schur), so the commutant is trivial and no proper invariant subspace can support its projection. The labels are read off by restricting to the two copies, so distinct pairs give inequivalent representations. ∎

So the Lorentz algebra's representation theory is a multiplication table: $(0,0)$ is the trivial representation; $(\frac{1}{2}, 0)$ and $(0, \frac{1}{2})$ have dimension two and are about to receive names; $(\frac{1}{2}, \frac{1}{2})$ has dimension four — a suspicious number over Minkowski space; $(1,0)$ and $(0,1)$ have dimension three.

Remark 14.7 (Real-form bookkeeping)**.** Three clauses keep the accounting honest. *(i)* Every representation of the real algebra $\mathfrak{so}(1,3)$ on a complex space extends uniquely complex-linearly to $\mathfrak{so}(1,3)_{\mathbb{C}}$ — Lecture 10's complexification lemma verbatim, with $\mathfrak{su}(2)$ replaced by $\mathfrak{so}(1,3)$ — so the pairs (j_1, j_2) classify the complex representations of the Lorentz algebra itself. *(ii)* Unlike the rotation story, complex conjugation acts: the conjugation of $\mathfrak{so}(1,3)_{\mathbb{C}}$ fixing the real form sends $\mathrm{i} \mapsto -\mathrm{i}$, hence $A_a \leftrightarrow B_a$, so the complex conjugate of (j_1, j_2) is (j_2, j_1). A Lorentz representation is equivalent to its conjugate only when $j_1 = j_2$: the algebra knows the difference between left and right. *(iii)* No group representation has yet been selected. Since $\mathrm{SO}^+(1,3)$ is not simply connected (Corollary 14.4), integration to that group is not automatic: Lecture 9 says to integrate uniquely on the universal cover and then test whether the covering kernel acts trivially. The cover and this descent test are two pages away.

14.4 $\mathrm{SL}(2,\mathbb{C})$, the double cover, and spinors

Lecture 11 promised that today would replay its construction one signature up; here is the replay. Encode an event $x \in \mathbb{R}^4$ as a Hermitian 2×2 matrix:

$$X = x^0 \mathbb{1} + \vec{x} \cdot \vec{\sigma} = \begin{pmatrix} x^0 + x^3 & x^1 - \mathrm{i}x^2 \\ x^1 + \mathrm{i}x^2 & x^0 - x^3 \end{pmatrix}, \qquad \det X = \eta(x, x) : \tag{14.5}$$

a linear bijection onto the Hermitian matrices, with the interval sitting in the determinant (in Lecture 11 the same trick put $|\vec{x}|^2$ there; the time component moved it to the Minkowski form). For $A \in \mathrm{SL}(2,\mathbb{C})$ the map $X \mapsto AXA^\dagger$ preserves Hermiticity and, since $|\det A|^2 = 1$, the determinant — so it defines a linear map $\Phi(A)$ of $\mathbb{R}^4$ preserving η: a homomorphism $\Phi\colon \mathrm{SL}(2,\mathbb{C}) \to \mathrm{O}(1,3)$, continuous because the entries of $\Phi(A)$ are quadratic in those of A. First, the topology of the source.

Theorem 14.8 (SL(2, ℂ) is simply connected)**.** *Polar decomposition (Sheet 8, complex case) factors every $A \in \mathrm{SL}(2,\mathbb{C})$ uniquely as $A = UP$ with U unitary and P positive-definite; here $\det P > 0$ and $|\det U| = 1$ force $\det P = \det U = 1$, so $U \in \mathrm{SU}(2)$, and $P = e^H$ for a unique traceless Hermitian H (positive matrices have unique Hermitian logarithms, and $\det e^H = e^{\operatorname{tr} H}$ — Lecture 8's formula — kills the trace). The resulting map*

$$\mathrm{SU}(2) \times \mathbb{R}^3 \longrightarrow \mathrm{SL}(2,\mathbb{C}), \qquad (U, H) \mapsto Ue^H,$$

is a homeomorphism, so $\mathrm{SL}(2,\mathbb{C}) \cong S^3 \times \mathbb{R}^3$: connected, and simply connected — a loop contracts factor by factor, on S^3 by Lecture 9's theorem and in $\mathbb{R}^3$ by convexity.

Exactly where $\mathrm{SO}^+(1,3)$ failed (Corollary 14.4), $\mathrm{SL}(2,\mathbb{C})$ succeeds: the boost factor $\mathbb{R}^3$ is topologically innocent in both, and the difference is entirely S^3 versus $\mathrm{SO}(3)$ — Lecture 11's gap, inherited intact.

Theorem 14.9 (The double cover of the Lorentz group)**.** *Φ maps $\mathrm{SL}(2,\mathbb{C})$ onto $\mathrm{SO}^+(1,3)$ with kernel $\{\pm\mathbb{1}\}$, so*

$$\mathrm{SO}^+(1,3) \;\cong\; \mathrm{SL}(2,\mathbb{C})/\{\pm\mathbb{1}\}.$$

Its derivative $\varphi(a)\colon X \mapsto aX + Xa^\dagger$ is an isomorphism of real Lie algebras $\mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}} \to \mathfrak{so}(1,3)$, with

$$\varphi(X_a) = L_a, \qquad \varphi\left(\tfrac{\sigma_a}{2}\right) = K_a \qquad (X_a = -\mathrm{i}\sigma_a/2 \text{ as always}).$$

Proof. Image in SO^+*:* $\mathrm{SL}(2,\mathbb{C})$ is connected, Φ continuous and $\Phi(\mathbb{1}) = \mathbb{1}$, so the image lies in the identity component. *Kernel:* one signature up, the same three lines as Lecture 11. If $AXA^\dagger = X$ for all Hermitian X, then $X = \mathbb{1}$ gives $AA^\dagger = \mathbb{1}$, so A is unitary and commutes with every Hermitian matrix, hence with all of $M_2(\mathbb{C})$, hence $A = c\mathbb{1}$ with $c^2 = \det A = 1$. *Surjectivity:* by the polar anatomy (Theorem 14.3) it suffices to hit all rotations and all boosts. Restricted to $\mathrm{SU}(2)$, A unitary, the action fixes $x^0\mathbb{1}$ and conjugates $\vec{x}\cdot\vec{\sigma}$: this is Lecture 11's double cover verbatim, so $\Phi(\mathrm{SU}(2))$ is all of $1 \oplus \mathrm{SO}(3)$, and its derivative on $\mathfrak{su}(2)$ is Lecture 11's $X_a \mapsto L_a$. For boosts take $A = e^{\chi\sigma_3/2} = \mathrm{diag}(e^{\chi/2}, e^{-\chi/2})$, Hermitian positive with determinant one: then $AXA^\dagger = AXA$ multiplies the diagonal of (14.5) by $e^{\pm\chi}$, i.e.

$$x'^0 \pm x'^3 = e^{\pm\chi}\,(x^0 \pm x^3), \qquad x'^{1,2} = x^{1,2},$$

which is precisely $B_3(\chi)$ of (14.1) — note the *halved rapidity* in A, the hyperbolic twin of Lecture 9's halved angle — and conjugating by $\mathrm{SU}(2)$ rotates the axis anywhere. Differentiating at $\chi = 0$ gives $\varphi(\sigma_3/2) = \frac{1}{2}\{\sigma_3, \cdot\} = K_3$ on coordinates. *Derivative an isomorphism:* the two halves above show φ hits the basis (14.3); for injectivity, $aX + Xa^\dagger = 0$ for all Hermitian X gives (at $X = \mathbb{1}$) $a^\dagger = -a$ and then $aX = Xa$ for all X, so a is a scalar, anti-Hermitian and traceless: zero. Dimensions $6 = 6$ finish it. ■

Two dividends, one per theorem. First, the projective story: by Wigner's theorem and Bargmann's (both as in Lecture 11, the latter imported there as a stated black box — $\mathfrak{sl}(2,\mathbb{C})$ semisimple, so $H^2 = 0$), every projective unitary representation of $\mathrm{SO}^+(1,3)$ comes from an honest representation of its simply connected double cover — the Lorentz group has the same luck as the rotation group, as Lecture 11 promised it would. Quantum relativity is the representation theory of $\mathrm{SL}(2,\mathbb{C})$. Second, the licence Lecture 10 left uncashed is finally spent:

Corollary 14.10 (The lifting theorem, consumed at last)**.** $\mathrm{SL}(2,\mathbb{C})$ *is simply connected, so by Lecture 9's lifting theorem every representation of its Lie algebra integrates uniquely to the group. In particular each* (j_1, j_2) *of Theorem 14.6 — a representation of* $\mathfrak{so}(1,3) \cong \mathfrak{sl}(2,\mathbb{C})_\mathbb{R}$ *with, until this moment, no group in sight — defines a representation* $\Pi_{(j_1,j_2)}$ *of* $\mathrm{SL}(2,\mathbb{C})$*. It descends to* $\mathrm{SO}^+(1,3)$ *exactly when* $\Pi_{(j_1,j_2)}(-\mathbb{1}) = +\mathbb{1}$*, and running Lecture 10's sign-flip computation on the total rotation generator* $L_3 = A_3 + B_3$ *gives* $\Pi_{(j_1,j_2)}(-\mathbb{1}) = (-1)^{2(j_1+j_2)}\,\mathbb{1}$*: integer* $j_1 + j_2$ *descends, half-integer* $j_1 + j_2$ *is projective — rays again.*

Names, at last. The *fundamental* representation of $\mathrm{SL}(2,\mathbb{C})$ — A acting on $\mathbb{C}^2$ as itself — kills the B-copy and is spin-$\frac{1}{2}$ for the A-copy (trace the definitions through φ: A_a pulls back to X_a, B_a to 0), so it is $(\frac{1}{2}, 0)$: its vectors $\psi_L \mapsto A\psi_L$ are *left-handed Weyl spinors*. Its complex conjugate $A \mapsto \bar{A}$ is $(0, \frac{1}{2})$ by Remark 14.7 (ii): *right-handed Weyl spinors*. The same representation is also realised by $A \mapsto (A^\dagger)^{-1}$, and the intertwiner between the two models is an old friend: Lecture 12's invariant ϵ-tensor, through the cofactor identity $\bar{A}^\mathsf{T}\epsilon\bar{A} = \epsilon$, gives $\epsilon\bar{A}\epsilon^{-1} = (\bar{A}^\mathsf{T})^{-1} = (A^\dagger)^{-1}$. (The physicists' dotted and undotted spinor indices are bookkeeping for the two copies and for ϵ's index-raising; we shall not need the apparatus, only the idea. Which copy is called "left" is convention; ours is fixed by $\psi_L \mapsto A\psi_L$.)

Remark 14.11 (Parity swaps the copies)**.** Conjugation by the parity $P = \eta$ fixes the rotation generators and reverses the boosts, $\mathrm{Ad}_P\colon L_a \mapsto L_a$, $K_a \mapsto -K_a$ (the K's straddle the time–space blocks), hence swaps $A_a \leftrightarrow B_a$: a parity transformation turns a (j_1, j_2) into a (j_2, j_1). A quantum theory of a single Weyl spinor therefore *cannot* represent parity; the smallest parity-symmetric spinor is

$$(\tfrac{1}{2}, 0) \oplus (0, \tfrac{1}{2}) :$$

the *Dirac spinor*, four components not because spacetime has four dimensions but because left and right each need two. (Nature, through the weak interaction, actually uses the unpaired option — parity violation is the experimental discovery that the two copies are physically distinct.)

Proposition 14.12 (Four-vectors are $(\frac{1}{2}, \frac{1}{2})$)**.** *The complexification of the defining representation of* $\mathrm{SO}^+(1,3)$*, pulled back along* Φ*, is* $\Pi_{(\frac{1}{2},\frac{1}{2})}$ *— and the dictionary* (14.5) *is itself the intertwiner.*

Proof. Pulled back, the representation is $X \mapsto AXA^\dagger$ on Hermitian matrices; its complexification is the same formula on all of $M_2(\mathbb{C})$. The map $\psi \otimes \bar{\chi} \mapsto \psi\chi^\dagger$ is an isomorphism $\mathbb{C}^2 \otimes \overline{\mathbb{C}^2} \to M_2(\mathbb{C})$ intertwining $A \otimes \bar{A}$ with $M \mapsto AMA^\dagger$ (check on rank ones: $A(\psi\chi^\dagger)A^\dagger = (A\psi)(A\chi)^\dagger$), so the representation is $(\frac{1}{2}, 0) \otimes (0, \frac{1}{2})$. Tensor products multiply slotwise — $(j_1, j_2) \otimes (j_1', j_2')$ decomposes by Lecture 12's Clebsch–Gordan series applied in each copy separately, since the A-operators act only on the first labels and the B-operators only on the second — and here each slot is $\frac{1}{2} \otimes 0$ or $0 \otimes \frac{1}{2}$: a single piece, $(\frac{1}{2}, \frac{1}{2})$, irreducible of dimension four. ∎

Remark 14.13 (Where electromagnetism sits)**.** By the slotwise rule, $(\frac{1}{2}, \frac{1}{2}) \otimes (\frac{1}{2}, \frac{1}{2}) = (0,0) \oplus (1,0) \oplus (0,1) \oplus (1,1)$, of dimensions $1+3+3+9 = 16$; the antisymmetric part has dimension six and is

$$\Lambda^2(\tfrac{1}{2}, \tfrac{1}{2}) = (1,0) \oplus (0,1) :$$

the seat of the electromagnetic field strength $F_{\mu\nu}$. The two halves are the self-dual and anti-self-dual fields, the combinations $\vec{E} \pm \mathrm{i}\vec{B}$ — electromagnetism decomposed by the Lorentz group before a single field equation is written, and another instance of conjugation (and parity) swapping (j_1, j_2) with (j_2, j_1).

14.5 The no-go theorem, and fields versus states

Quantum mechanics wants unitary representations; the (j_1, j_2) are not unitary — the defining representation already shows $\mathrm{i}K_a$ anti-Hermitian where unitarity would demand Hermitian. Compactness gives useful intuition, but noncompactness alone is no obstruction: the noncompact group $\mathbb{R}$ has nontrivial one-dimensional unitary characters $t \mapsto e^{\mathrm{i}\lambda t}$. What rules out the Lorentz representations at hand is the exact Lorentz bracket structure together with anti-Hermiticity and the finite-dimensional trace argument. Here is that algebraic proof.

Theorem 14.14 (No finite-dimensional unitary representations)**.** *Every finite-dimensional unitary representation of* $\mathrm{SO}^+(1,3)$ *— equivalently of* $\mathrm{SL}(2,\mathbb{C})$ *— is trivial.*

Proof. Let Π be unitary on a finite-dimensional V, π its derivative. By Lecture 9's workhorse, $\pi(L_a)$ and $\pi(K_a)$ are all anti-Hermitian. Extend complex-linearly and set $\mathcal{A}_a = \frac{1}{2}\big(\pi(L_a) - \mathrm{i}\pi(K_a)\big)$, $\mathcal{B}_a = \frac{1}{2}\big(\pi(L_a) + \mathrm{i}\pi(K_a)\big)$, which inherit the decoupled brackets of Theorem 14.5. Anti-Hermiticity of the generators gives

$$\mathcal{A}_a^\dagger = \tfrac{1}{2}\big(-\pi(L_a) - \mathrm{i}\pi(K_a)\big) = -\mathcal{B}_a,$$

so the two commuting copies are each other's adjoints — the relativistic twist with no $\mathfrak{su}(2)$ analogue. In particular $[\mathcal{A}_a, \mathcal{A}_b^\dagger] = -[\mathcal{A}_a, \mathcal{B}_b] = 0$. Now the trace trick: from $\mathcal{A}_3 = [\mathcal{A}_1, \mathcal{A}_2]$,

$$\mathrm{tr}\big(\mathcal{A}_3\mathcal{A}_3^\dagger\big) = \mathrm{tr}\big(\mathcal{A}_1\mathcal{A}_2\mathcal{A}_3^\dagger - \mathcal{A}_2\mathcal{A}_1\mathcal{A}_3^\dagger\big) = \mathrm{tr}\big(\mathcal{A}_2\,[\mathcal{A}_3^\dagger, \mathcal{A}_1]\big) = 0,$$

cycling the first term and using that the daggered copy commutes. But $\mathcal{A}_3\mathcal{A}_3^\dagger$ is positive semidefinite, so $\mathcal{A}_3 = 0$, and relabelling axes kills every $\mathcal{A}_a$, hence every $\mathcal{B}_a = -\mathcal{A}_a^\dagger$, hence

$\pi(L_a) = \mathcal{A}_a + \mathcal{B}_a = 0$ and $\pi(K_a) = \mathrm{i}(\mathcal{A}_a - \mathcal{B}_a) = 0$. All generators vanish, and on a connected group the representation is trivial. ■

Read the two classifications side by side and the finale's organising principle appears. *Finite-dimensional, non-unitary representations classify fields: the* (j_1, j_2) *— scalars, Weyl and Dirac spinors, vectors, field strengths — are the transformation types a relativistic field can have. Infinite-dimensional unitary representations classify states: particles.* That is Lecture 15's program in one sentence. There is no paradox in the pairing: a field is a bookkeeping device, free to transform non-unitarily because no probability lives in its components; probability lives in the state space, which Theorem 14.14 forces to be infinite-dimensional — and this is precisely why Lecture 13 normalised infinite-dimensional unitary representations before relativity made them compulsory. (Infinite-dimensional unitary representations of the Lorentz group alone do exist — the principal series, starred further reading — but physics will want the full Poincaré group, translations included.)

14.6 The Dirac equation from representation theory (starred)

Starred: examinable only through Sheet 14. Write, for a momentum $p \in \mathbb{R}^4$,

$$\sigma(p) = p^0 \mathbb{1} + \vec{p} \cdot \vec{\sigma}, \qquad \bar{\sigma}(p) = p^0 \mathbb{1} - \vec{p} \cdot \vec{\sigma},$$

the first being (14.5), the second its cofactor companion: $\epsilon\, \sigma(p)^{\mathsf{T}} \epsilon^{-1} = \big(\operatorname{tr} \sigma(p)\big)\mathbb{1} - \sigma(p) = \bar{\sigma}(p)$. The first transforms by definition of Φ; the second's law follows by transposing and conjugating with ϵ, using the cofactor identity twice:

$$\sigma(\Lambda p) = A\, \sigma(p)\, A^{\dagger}, \qquad \bar{\sigma}(\Lambda p) = (A^{\dagger})^{-1}\, \bar{\sigma}(p)\, A^{-1}, \qquad \Lambda = \Phi(A).$$

Now let ψ_L transform as $(\frac{1}{2}, 0)$ and ψ_R as $(0, \frac{1}{2})$ in the $(A^{\dagger})^{-1}$ model. The laws above say that $\bar{\sigma}(p)$ *eats left and produces right*: $\bar{\sigma}(\Lambda p)\, A\psi_L = (A^{\dagger})^{-1}\big(\bar{\sigma}(p)\psi_L\big)$, and symmetrically $\sigma(p)$ eats right and produces left. The only linear, momentum-covariant way to couple the two Weyl halves with a scale m is therefore the pair

$$\bar{\sigma}(p)\, \psi_L = m\, \psi_R, \qquad \sigma(p)\, \psi_R = m\, \psi_L, \tag{14.6}$$

and its consistency condition is the physics: applying the two maps in succession, $\sigma(p)\bar{\sigma}(p) = \big((p^0)^2 - |\vec{p}|^2\big)\mathbb{1} = \eta(p, p)\, \mathbb{1}$, so nonzero solutions exist only on the mass shell $\eta(p, p) = m^2$ — the relativistic dispersion relation as an integrability condition. Stack $\psi = (\psi_L, \psi_R)$ on the Dirac space $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$ and define

$$\gamma^0 = \begin{pmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0 \end{pmatrix}, \qquad \gamma^k = \begin{pmatrix} 0 & -\sigma_k \\ \sigma_k & 0 \end{pmatrix}, \qquad \text{so that} \qquad \gamma^\mu p_\mu = \begin{pmatrix} 0 & \sigma(p) \\ \bar{\sigma}(p) & 0 \end{pmatrix}$$

(with $p_\mu = \eta_{\mu\nu} p^\nu$; conventions for the γ's vary across books exactly as the left/right convention does — ours is pinned by $\psi_L \mapsto A\psi_L$). Then (14.6) is one equation, $\gamma^\mu p_\mu \psi = m\psi$, or in position space

$$\big(\mathrm{i}\gamma^\mu \partial_\mu - m\big)\psi = 0\ :$$

the *Dirac equation* — not an equation that happens to be covariant, but the unique first-order intertwiner between the two Weyl halves, with the mass shell as its integrability condition. Block multiplication verifies

$$\{\gamma^\mu, \gamma^\nu\} = 2\, \eta^{\mu\nu}\, \mathbb{1}, \tag{14.7}$$

the Clifford algebra of spacetime: Lecture 13's fermionic nursery — the γ's built there from creation and annihilation operators with $\delta_{\mu\nu}$ — now grown, with the Euclidean δ replaced by η. Sheet 14 checks that $\frac{1}{4}[\gamma^\mu, \gamma^\nu]$ represents $\mathfrak{so}(1,3)$ and reassembles $(\frac{1}{2},0) \oplus (0,\frac{1}{2})$. Two parting observations. At $m = 0$ the pair (14.6) decouples and a single Weyl spinor suffices — a massive fermion needs both halves (or must marry its own conjugate: Majorana, further reading). And Remark 14.11 now has teeth: mass and parity each demand the doubled spinor, which is why Dirac's electron has four components.

Here the kinematics of the finale stands complete: the group of spacetime, its double cover, the fields it allows, and the theorem that evicts its states from finite dimensions. Lecture 15 adds the translations — the Poincaré group — and carries out the program announced above: the infinite-dimensional unitary irreducible representations, constructed by Wigner's method, are the particles, labelled by mass and spin. The course ends where physics begins.

Remark 14.15 (Computer algebra)**.** Notebook 14 implements Φ symbolically from (14.5) and verifies the double cover numerically (A and $-A$ landing on the same Λ); composes two perpendicular boosts, polar-decomposes the product by Theorem 14.3, and plots the Thomas–Wigner rotation angle against the two rapidities; tabulates (j_1, j_2) dimensions against the field content of electrodynamics; and checks the Clifford relations (14.7) by brute matrix multiplication.

Summary.

- *Lorentz group.* $\mathrm{O}(1,3)$ has four connected components (Proposition 14.2); the restricted group factors as a rotation times a boost (Theorem 14.3).
- *Boost algebra.* $[K_i, K_j] = -\epsilon_{ijk}L_k$ (eq. (14.3)): boosts do not close, the seed of Thomas precession.
- *Decoupling.* $\mathfrak{so}(1,3)_{\mathbb{C}} \cong \mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$ (Theorem 14.5); irreducibles are labelled (j_1, j_2) (Theorem 14.6), with four-vectors at $(\frac{1}{2},\frac{1}{2})$ (Proposition 14.12).
- *Double cover.* $\mathrm{SL}(2,\mathbb{C})$ is simply connected (Theorem 14.8) and double-covers the restricted Lorentz group (Theorem 14.9); its $(\frac{1}{2},0)$ and $(0,\frac{1}{2})$ are the Weyl spinors.
- *No-go.* The Lorentz group has no non-trivial finite-dimensional unitary representation (Theorem 14.14): fields are finite-dimensional but non-unitary, states unitary but infinite-dimensional.

Tutorial. Exercise Sheet 14.

References

Hall §2.5 (polar decomposition), Ch. 1 ($\mathrm{O}(1,3)$); Woit (the chapters on Minkowski space and on the Lorentz group and its representations); Costa–Fogli Ch. 3–5; Tung (the Lorentz group); Schwichtenberg Ch. 3 (motivation-level parallel reading); Jones Pt. 11.

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